

18th March, 26 Wed

Assignment on linear system formulation

① Resource allocation in cloud CPU task scheduling (CSE)

(a) Decision variables vector consists of

$$\vec{p} = \begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix} = \begin{pmatrix} n(\text{Data processing}) \\ n(\text{Rendering}) \\ n(\text{Encryption}) \end{pmatrix}$$

where $n(\text{process})$ gives the number of processes scheduled.

X (b) From the resource - process matrix/table is given by $\tilde{S}$, (scheduler) a transformation

$$\tilde{S}: \text{process space} \rightarrow \text{resource space}$$

The matrix representation of $\tilde{S}$ is given by action of $\tilde{S}$ on the resource process basis vectors, $\{\vec{r}_1, \vec{r}_2, \vec{r}_3\}$

where $\vec{r}_1$ is unit vector along RAM axis

$\vec{r}_2$ is unit vector along CPU axis

$\vec{r}_3$ is unit vector along I/O axis

Since RAM, CPU, and I/O resources, are independent of each other, if the resource basis vectors are considered as computational basis vectors, then

$$\vec{r}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \vec{r}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{r}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Action of scheduler:

$$\tilde{S} \vec{r}_1 = \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix}, \quad \tilde{S} \vec{r}_2 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \quad \tilde{S} \vec{r}_3 = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$$

so that
$$\tilde{S} = \begin{pmatrix} \tilde{S} \vec{r}_1 & \tilde{S} \vec{r}_2 & \tilde{S} \vec{r}_3 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 1 & 3 \\ 4 & 2 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

Resource constraints:

As a system of linear equations,

There are resource constraints on number of RAM, CPU and I/O units.

X (i) RAM constraint

If

(b) The scheduler $\tilde{S}$ can be modelled as a transformation from the process space to the resource space. If $\tilde{S}$ is the matrix representation of the scheduler, then

$$\tilde{S} : \mathcal{P}_p \rightarrow \mathcal{P}_R$$

where $\mathcal{P}_p$ is the process space and $\mathcal{P}_R$ is the resource space

Resource constraints.

Since the system is limited, there are resource constraints.

(i) RAM constraint.

If the process space can be modelled as linear space spanned by 3 independent unit vectors $\{\vec{P}_1, \vec{P}_2, \vec{P}_3\} = (\hat{DATA}, \hat{RENDER}, \hat{I/O})$

Since every data processing task requires 2 RAM units
rendering process requires 4 RAM units
Encryption process requires 1 RAM unit

The total number of RAM units required are

$$n(RAM) = 2P_1 + 4P_2 + 1P_3$$

$$\text{Given } n(RAM) = 40 \text{ (GB)}$$

$$\text{So } 2P_1 + 4P_2 + P_3 = 40$$

(ii) CPU constraint

Similarly,

Data processing requires	1 CPU unit
Rendering requires	2 CPU units
I/O Encryption requires	2 CPU units

So total number of CPU units required are

$$n(CPU) = 1P_1 + 2P_2 + 2P_3$$

$$\text{Given } n(CPU) = 20$$

$$\text{So } P_1 + 2P_2 + 2P_3 = 20$$

(iii) I/O constraint

Similarly

$$3P_1 + 2P_2 + P_3 = 30$$

So the system of linear equations is given by

$$2p_1 + 4p_2 + p_3 = 40$$

$$p_1 + 2p_2 + 2p_3 = 20$$

$$3p_1 + 2p_2 + p_3 = 30$$

B (C) Matrix representation of system of linear equations is

$$\tilde{A} \vec{x} = B \equiv \tilde{S} \vec{p} = \vec{r}$$

Here $\vec{r}$ is the resource vector in the resource space S_R .

S_R is spanned by 3 independent vectors

$$\{\vec{r}_1, \vec{r}_2, \vec{r}_3\} = (RAM, CPU, IO)$$

$$\underbrace{\begin{pmatrix} 2 & 4 & 1 \\ 1 & 2 & 2 \\ 3 & 2 & 1 \end{pmatrix}}_{\tilde{S}} \underbrace{\begin{pmatrix} p_1 \\ p_2 \\ p_3 \end{pmatrix}}_{\vec{p}} = \underbrace{\begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix}}_{\vec{r}} \quad \begin{matrix} \tilde{A} \vec{x} = \vec{b} \\ \parallel \\ \tilde{S} \vec{p} = \vec{r} \end{matrix}$$

(d) Here, in the matrix representation, the process vector $\vec{p}$ is a 3-column vector
resource vector $\vec{r}$ is a 3-column vector

and scheduler

$\tilde{S}$ is a 3×3 square matrix

Multiple allocations of resources are possible if the solutions are multiple. The number of solutions is ~~is~~ determined by the rank of the scheduler matrix $RANK(\tilde{S})$. If $RANK(\tilde{S}) < 3$, then infinite solutions

(c) For a unique solution, the $\text{RANK}(\tilde{S}) = 3$

Extension:

If there is additional constraint of total number of ~~tasks~~ _{processed} equal to 15.

$$\text{then } P_1 + P_2 + P_3 = 15$$

The system of linear equations has 4 linear equations in 3 variables/unknowns so, the scheduler matrix is given by

$$\begin{pmatrix} 2 & 4 & 1 \\ 1 & 2 & 2 \\ 3 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}_{4 \times 3} \cdot \begin{pmatrix} P_1 \\ P_2 \\ P_3 \end{pmatrix}_{3 \times 1} = \begin{pmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \end{pmatrix}_{4 \times 1}$$

Now the scheduler is modified and also the resource space.

$$\tilde{S}' : \mathcal{S}_{R'} \rightarrow \mathcal{S}_P$$

Here $\mathcal{S}_{R'}$ is the modified resource space

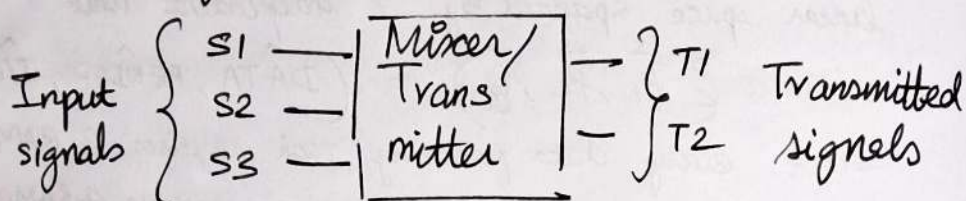
The scheduler is no longer square matrix

For a unique solution,

$$\text{RANK}(\tilde{S}') = 3$$

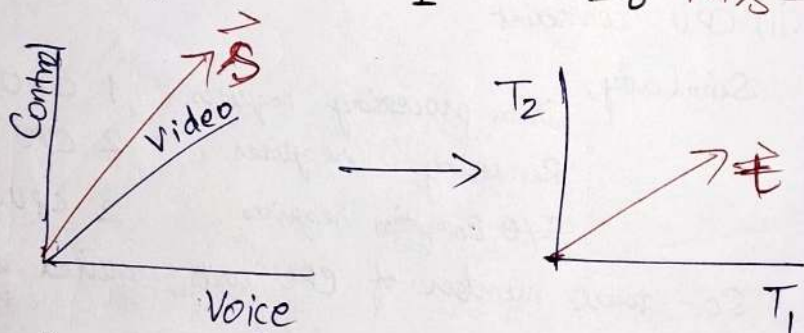
② Signal mixing in a communication transmitter (ECE)

This problem is called linear signal mixing



The ~~transmitter~~ ^{mixer} ($\tilde{M}$) can be modelled as a linear transformation from the input signal space ($\mathcal{S}_I$) to the output transmitted space ($\mathcal{S}_O$)

$$\tilde{M} : \mathcal{S}_I \longrightarrow \mathcal{S}_O \quad \tilde{M} \vec{s} = \vec{t}$$



input signal space $\rightarrow$ transmitted space

- The signal variables are voice, video, and control signals
- Inside the mixer, the input signals are amplified with different gains, then added and transmitted

The different gains are coefficients in the matrix

$$T_1 = 2S_1 + 1S_2 + 3S_3$$

$$T_2 = 1S_1 + 4S_2 + 2S_3$$

$$\Rightarrow \begin{aligned} 2S_1 + 1S_2 + 3S_3 &= T_1 \\ 1S_1 + 4S_2 + 2S_3 &= T_2 \end{aligned}$$

$$\Rightarrow \underbrace{\begin{pmatrix} 2 & 1 & 3 \\ 1 & 4 & 2 \end{pmatrix}}_{\tilde{M} \quad 2 \times 3} \underbrace{\begin{pmatrix} S_1 \\ S_2 \\ S_3 \end{pmatrix}}_{\vec{S} \quad 3 \times 1} = \underbrace{\begin{pmatrix} T_1 \\ T_2 \end{pmatrix}}_{\vec{T} \quad 2 \times 1}$$

$\tilde{A} \vec{x} = \vec{b}$
 $\parallel$
 $\therefore \tilde{M} \vec{S} = \vec{T}$

(C) Miner $\tilde{M}$ has dimensions 2×3

Input signal $\vec{S}$ has dimension 3×1

Transmitted signal $\vec{T}$ has dimension 2×1

We have a system of 2 linear equations in 3 variables.

Since $n(\text{equations}) < n(\text{variables})$, the linear system is an under-determined.

For unique recovery of ^{input} signal,

$$\text{RANK}(\tilde{M}) = 3$$

However, that is impossible!

Case (i)

(d) If $\text{RANK}(\tilde{M}) < 2, \Rightarrow \text{RANK}(\tilde{M}) = 1$
 then only one independent equation exists. So infinitely many solutions exist and they lie on a plane in the input signal space.

Case (ii)

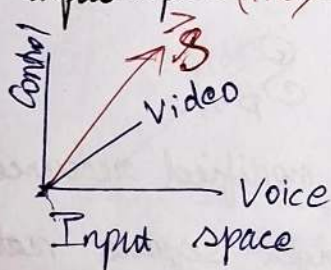
If no solution exists, then $\text{RANK}(\tilde{M}) = 0$.
 The two linear equations represent parallel planes that never intersect.

Extension:

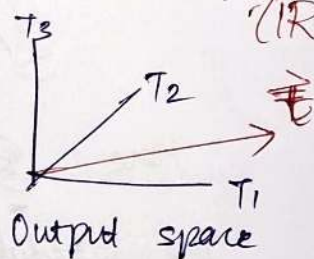
Now there are 3 output and 3 input signals



So the transformation is from 3 dimensional input space ($\mathbb{R}^3$) to 3 dimensional output space ($\mathbb{R}^3$)



miner $\rightarrow$



$$\tilde{M}' : \mathcal{S}_I \rightarrow \mathcal{S}_O \quad \tilde{M}' \vec{S} = \vec{T}$$

$$\mathbb{R}^3 \rightarrow \mathbb{R}^3$$

Here $\mathbb{R}$ stands for real number set type of and superscript 3 stands for set of three real value coordinates.
 $\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R}$

The matrix dimension of miner is 3×3

If the $\text{RANK}(\tilde{M}') = 3$, then solution is unique.

Concept of RANK of a matrix determines the uniqueness of solution.

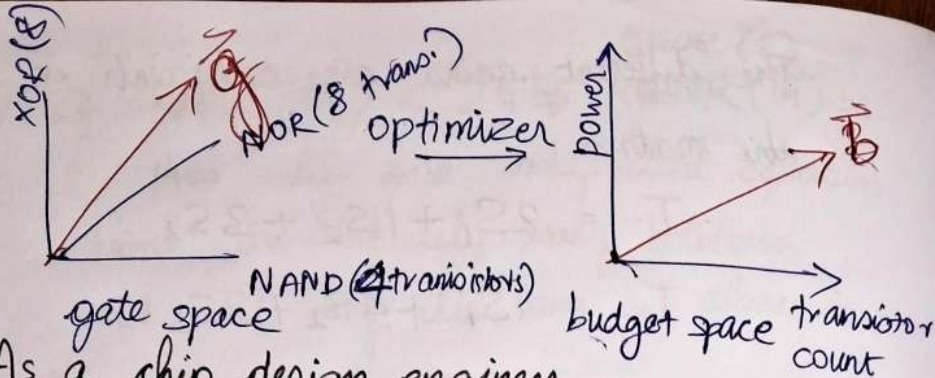
③ Transistor budget optimization in VLSI chip design (ECE $\rightarrow$ VLSI)

As a design engineer, we need to play in the design gate space that is spanned by the NAND, NOR, and XOR gates.

However, as an engineer, we need to respect constraints. These constraints can be modelled as a budget vector in the budget space, that is spanned by the number of transistors and thermal available power.

Number of transistors determine the form factor of the chip.

Available power sets upper bound on the ~~power~~ thermal losses or heat dissipation that determines the battery limitation and chip cooling ~~issue~~ requirements.



As a chip design engineer,
we need to determine the design specifications
keep in mind the engineering constraints



$$\tilde{O} : \mathcal{S}_G \rightarrow \mathcal{S}_B \quad \tilde{O} \vec{g} = \vec{b}$$

(a) The gate variables are $n(\text{NAND})$, $n(\text{NOR})$,
and $n(\text{XOR})$

(b) Transistor constraint = 4000

If $G1$ NAND, $G2$ NOR, and $G3$ XOR
gates are used, then

$$4G1 + 4G2 + 8G3 = 4000$$

(c) Power constraint = 2500 mW

Each NAND uses 2 mW

NOR uses 3 mW

XOR uses 5 mW

then

$$2G_1 + 3G_2 + 5G_3 = 2500$$

(d) $4G_1 + 4G_2 + 8G_3 = 4000$

$$2G_1 + 3G_2 + 5G_3 = 2500$$

$$\tilde{A} \vec{x} = \vec{b} \Rightarrow \begin{pmatrix} 4 & 4 & 8 \\ 2 & 3 & 5 \end{pmatrix} \begin{pmatrix} G_1 \\ G_2 \\ G_3 \end{pmatrix} = \begin{pmatrix} 4000 \\ 2500 \end{pmatrix}$$

2×3 3×1 2×1

(e) Optimizer $\tilde{O}$ has dimensions 2×3
 Gate vector $\vec{g}$ has dimension 3×1
 Budget vector $\vec{b}$ has dimension 2×1

(f) Not square
 Under-determined

(g) Yes multiple solutions possible.

Extension:

If additional constraint of $n(\text{XOR}) = 200$
 $\Rightarrow G_3 = 200$

we have system of 3 linear equations in 3 variables.

So system is square
 optimizable

$\text{RANK}(\tilde{O}) = 3 \Rightarrow$ unique solution

④ Gene expression modeling in cellular pathway (B.T.)

Gene $\rightarrow$ mRNA $\rightarrow$ Protein



The process of transcription takes a gene vector in the gene space S_G and

transforms it into a protein vector in the protein space $\mathcal{S}_P$

$$\gamma: \mathcal{S}_G \rightarrow \mathcal{S}_P$$

(A) The unknown variables are the genes G_1, G_2, G_3

The higher the gene activity, the more production of protein

$$\text{So } 3G_1 + 2G_2 + 1G_3 = 50$$

$$1G_1 + 4G_2 + 2G_3 = 60$$

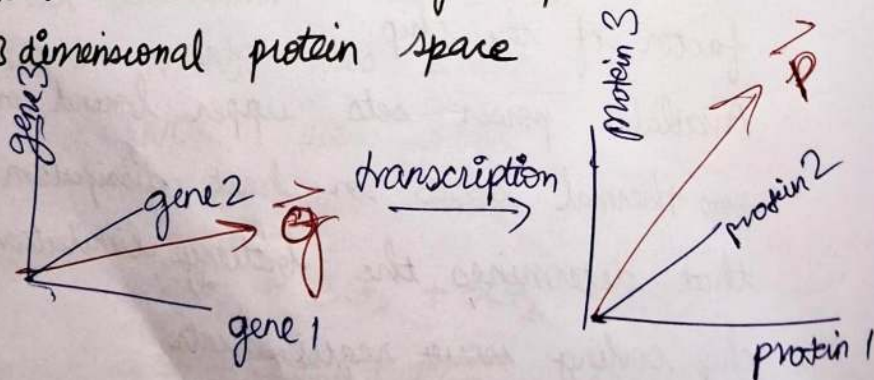
$$2G_1 + 1G_2 + 3G_3 = 55$$

Here we have 3 proteins under consideration

~~So~~ P_1, P_2, P_3

So in the production of P_1 , G_1 has more gene activity, however there is a contribution from the other two genes.

Similarly for P_2 & P_3 . This process can be modelled ~~visualized~~ visualized as mapping between 3-dimensional gene space to 3-dimensional protein space



$$(b) \begin{pmatrix} 3 & 2 & 1 \\ 1 & 4 & 2 \\ 2 & 1 & 3 \end{pmatrix} \begin{pmatrix} G_1 \\ G_2 \\ G_3 \end{pmatrix} = \begin{pmatrix} 50 \\ 60 \\ 55 \end{pmatrix}$$

$\underbrace{\hspace{10em}}_{3 \times 3} \quad \underbrace{\hspace{5em}}_{3 \times 1} \quad \underbrace{\hspace{5em}}_{3 \times 1}$
 Dimensions of $\vec{T} \cdot \vec{g} = \vec{p}$

$$\therefore \vec{T} \cdot \vec{g} = \vec{p}$$

$$A \vec{x} = \vec{b}$$

(c) Dimensions of transcripter $\vec{T}$ are 3×3

Dimensions of gene vector $\vec{g}$ are 3×1

Dimensions of protein vector $\vec{p}$ are 3×1

Transcripter is square matrix

Unique solution exists $\Leftrightarrow \text{RANK}(\vec{T}) = 3$

(d) If $\det(\vec{T}) = 0$,

there is no solution to $\vec{T} \vec{g} = \vec{p}$

$\Rightarrow$ No Gene activity is insufficient to produce desired results of proteins

$\text{RANK}(\vec{T}) < 3 \Rightarrow$ infinitely many solutions are possible

Extension:

For 5 proteins, and 3 genes

the transcripter has dimensions

there are 5 equations in 3 unknowns.

